

Manufacturer's installation manual shall be submitted and followed for full compliance. All insulation work shall be carried out by skilled workmen specially trained and certified by manufacturer in this kind of work. All insulated pipes shall be labeled (S.R. or R.R.) and provided with 300 mm wide band of paint along circumference at every 1200 mm for colour coding. Direction of fluid shall also be marked. Un-insulated MS pipes shall be painted throughout and direction of fluid marked. All painting shall be as per relevant BIS codes.

7.5 Protective Coating / Vapor Barrier Over Insulation

All ducts and pipes (On the roof / outside) exposed to UV rays shall be covered with two coats of UV paint / epoxy.

The insulation shall be continuous over the entire run of piping, fittings and valves.

Insulation shall be applied only after the piping system has been satisfactorily tested for leaks at 2 times the working pressure or at minimum 10 kg/sq.cm. test pressure. Insulation shall be covered with 26 gauge GI sheet cladding as per Schedule of Quantities and finished in neat and clean manner so as to achieve true surface. All longitudinal and transverse joints in the outer cladding shall have a minimum overlap of 50 mm duly beaded and grooved and shall be sealed with elastomeric metal sealant 95-44 of Benjamin Foster USA, or equivalent. Use of screws for fastening may puncture vapor barrier hence GI bands 0.50mm thick x 25 mm wide shall be provided at every 500 mm to retain cladding in position. Adhesive component once opened shall be used immediately and no leftovers shall be permitted to be used following day.

All insulation work shall be carried out by skilled workmen specially trained in this kind of work. All insulated pipes shall be labeled (S, R or RR) and provided with band of paint for color coding as per IS codes. Direction of fluid shall also be marked. Un-insulated M S pipes shall be painted throughout, and direction of fluid marked. All painting shall be as per relevant BIS codes.

Insulated pipes exposed to UV rays shall be covered with fiberglass fabric. Over fabric one coat of fireproof or Epoxy acrylic compound shall be applied. The coat shall be allowed to cure to non – stick state. Subsequently second coat of compound shall be applied to give tough and smooth finish to the insulated surface.

7.6 Duct Acoustic Lining

Open Cell Nitrile
Rubber



Noted

Comply

Duct acoustic lining material shall be Nitrile Rubber open cell foam. Thermal conductivity of the insulation material shall not exceed $0.047 \text{ W/m}^2\text{K}$ at an average temperature of 20°C . Density of the nitrile rubber shall be $140 - 180 \text{ Kg/m}^3$. The material should withstand maximum surface temperature of $+85^\circ\text{C}$ and minimum surface temperature of -20°C . The material should conform to Class 1 rating for surface spread of Flame in accordance to BS 476 Part 7 & HBF, HF 1 & HF 2 in accordance to UL 94, 1996.

Insulation should have antimicrobial product protection and should pass Fungi Resistance as per ASTM G 21 and Bacterial Resistance as per ASTM E 2180. The insulation should pass Air Erosion Resistance Test in accordance to ASTM Standard C 1071-05 (section 12.7).

Thickness of the material shall be of thickness specified for the individual application and with noise absorption proprieties as per IS: 8225 / ISO 354 / ASTM423C. The insulation should be installed as per manufacturer's recommendation.

Ducts so identified and marked on drawings and included in Schedule of Quantities for acoustic lining shall be provided with this insulation material for a distance of minimum 5 meters.

Adhesive used for sealing of insulation shall be non-flammable and with low VOC content (less than 50 gm/l as per USGBC (LEED) or IGBC guidelines) strictly as per manufacturer's recommendations.

Acoustic lining in Plenums especially for Air diffusion connected to slot diffusers shall have insulation as specified above.

7.7

Comms room / UPS Room Floor Insulation

Floor of data center shall be insulated with 19mm thick aluminum foil faced nitrile rubber insulation as per specifications. 600mm x 600mm grid shall be made on the floor. (Insulation grid shall be of same size as of data center floor grid). This grid shall be made from 19 mm wide 22G GI C-section. These C-sections shall be screwed to floor. Depth of screwing into the floor shall not be more than 5mm. Sections of insulation shall be stuck to floor with self-adhesive / adhesive solution as recommended by insulation manufacturer.



7.8 Acoustic Lining Of Mechanical Rooms

Mechanical room walls and ceiling of air conditioning plant room and air handling unit / fan rooms may be provided with acoustic lining of resin bonded Fibre glass with factory laminated black glass tissue as per Schedule of Quantities and as shown on the Drawings. The surface shall be cleaned and framework of 22 gage GI fabricated Channels 25 mm x 50 mm screwed back to back at 60 cm centers shall be provided vertically and horizontally so that 60 x 60 cm squares are formed. The gaps between frames shall be filled with 50 mm thick about 60 cm x 60 cm cut panels of resin bonded Fibre glass slabs. The insulation shall then be covered with 26 gauge perforated aluminum sheet, 60 cm or 120 cm wide having at least 40-50 percent perforations, fixed with sheet metal screws. Over-lapping of sheets shall be covered with aluminum beading. Acoustic lining of walls shall be terminated approximately 15 cm above the finished floor to prevent damage to insulation due to accidental water-logging in plant/AHU rooms.

7.9 Underdeck Insulation For Roof

Underdeck insulation shall be either of the following with density greater than or equal to 25 Kg/m³ & thermal conductivity of 0.21 Btu in/ft² hr.° F (at 24° C as per ASTM C-518).

25mm thick Closed cell elastomeric nitrile rubber with factory finish aluminum foil protective coating.

Underdeck surface of ceiling shall be cleaned and made dirt free. Insulation panels as mentioned above shall be pasted on this surface with black CPRX compound / SR 998 rubber adhesive. 28g wire net shall be tightened around insulation so as to avoid any kind of sagging. Ends of net shall be overlapping by at least 25mm. Overlaps shall be screwed with galvanized screws to avoid rusting.

7.10 Sound / Cross talk Attenuators

Attenuators shall be installed in ducts in accordance with requirements of drawings and as included in Schedule of Quantities.

Noise levels within conditioned spaces shall be not greater than those set out in schedule below:



Noted